Basic Concepts of Cognitive Radio

RADIO SPECTRUM

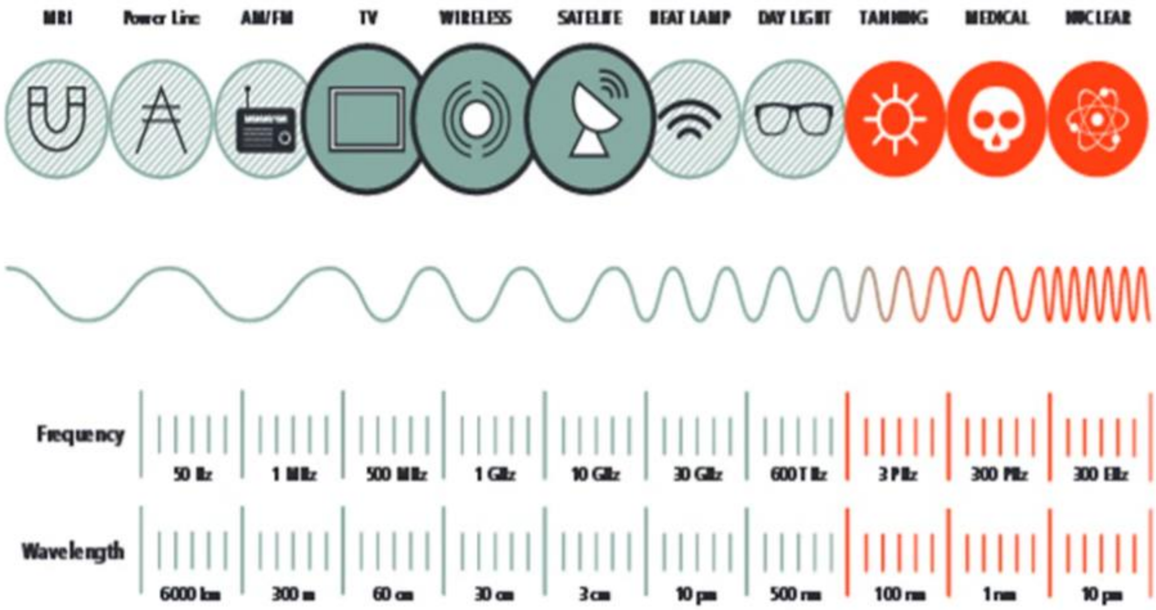
- The radio spectrum is the part of the electromagnetic spectrum from 3 Hz to 3000 GHz (3 THz). The electromagnetic waves in this frequency range are called radio waves, and are widely used in modern technology, particularly in telecommunication.
- To prevent interference between different users, the generation and transmission of radio waves is strictly regulated by national laws, coordinated by the international body, the International Telecommunication Union (ITU).
- Different parts of the radio spectrum are allocated by the ITU for different radio transmission technologies and applications.
- In some cases, parts of the radio spectrum are sold or licensed to operators of private radio transmission services (for example, cellular telephone operators or broadcast television stations).
- Range of allocated frequencies are often referred to by their provisioned use (for example, cellular spectrum or TV spectrum).

RADIO SPECTRUM

- The radio spectrum has been divided into decades of frequency range and each has its own designation as listed in Table.
- The radio spectrum is the range of frequencies used for wireless applications such as broadcast television and radio, cell phones, satellite radio, wireless computer networks, Bluetooth, GPS and countless other general and specialized applications that we use every day.

Frequency range	Designation	Abbreviation
3 – 30 KHz	Very Low Frequency	VLF
30 – 300 KHz	Low Frequency	LF
300 – 3000 KHz	Medium Frequency	MF
3 – 30 MHz	High Frequency	HF
30 – 300 MHz	Very High Frequency	VHF
300 – 3000 MHz	Ultra High Frequency	UHF
3 – 30 GHz	Super High Frequency	SHF
30 – 300 GHz	Extremely High Frequency	EHF

APPLICATIONS OF DIFFERENT FREQUENCY SPECTRUM



USERS IN CR

Primary User

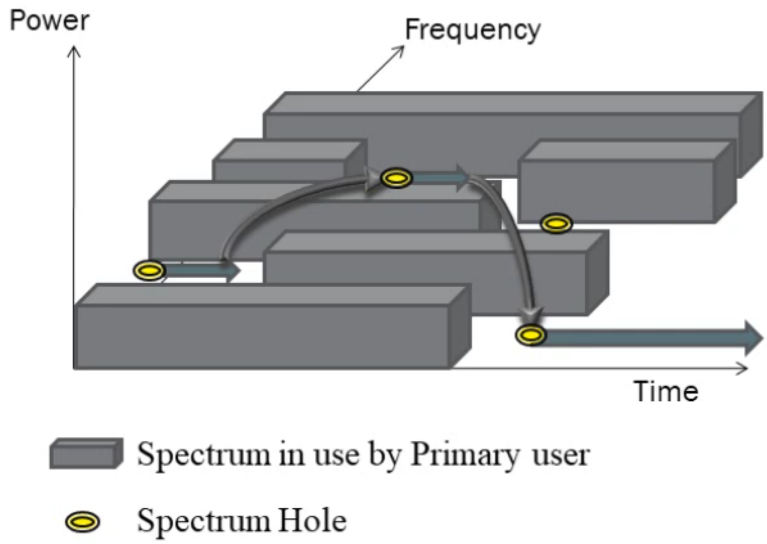
Licensed User
Highest priority

Secondary User

Unlicensed User
Cognitive-radio enabled users
Lower priority than PUs

SPECTRUM HOLE / WHITE SPACE

- A spectrum hole is a band of frequencies assigned to a primary user, but, at a particular time and specific geographic location, the band is not being utilized by that user.

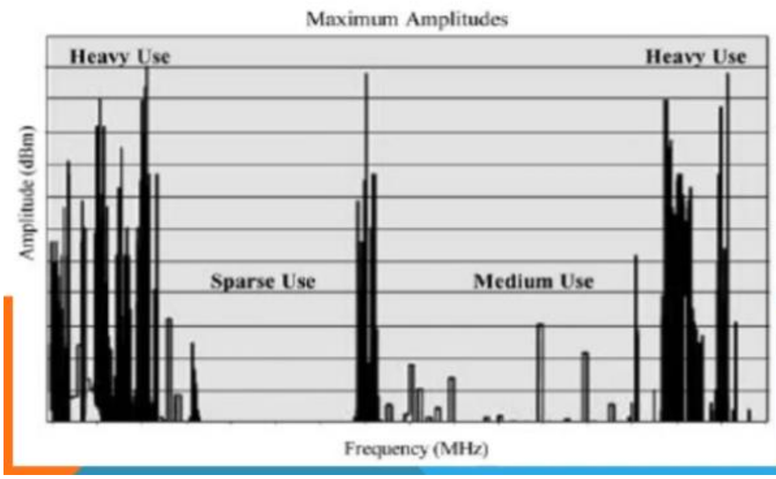


DYNAMIC SPECTRUM ACCESS

- DSA is a promising technology to alleviate the spectrum scarcity problem and increase spectrum utilization.
- Dynamic spectrum access is a new paradigm that allows secondary users (unlicensed users) to access the spectrum holes or white spaces in the licensed spectrum bands.
- It is the most vital and major commercial application of CR.
- **There are three main functions in Dynamic Spectrum Access:**
  i) Spectrum awareness
  ii) cognitive processing, and
  iii) spectrum access.
- **The main task of DSA is to overcome two types of interference:**
  - Harmful interference caused by device malfunctioning.
  - Harmful interference caused by malicious user.


WHAT IS COGNITIVE RADIO?
WHY WE NEED CR?

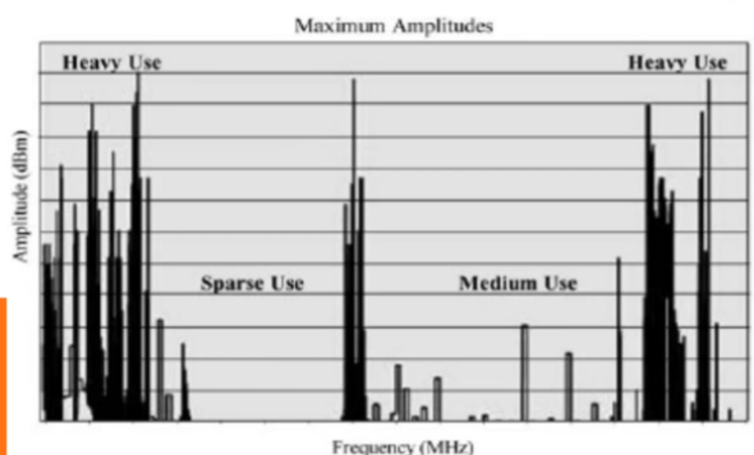
SPECTRUM FACTS



SPECTRUM FACTS

- Fixed Spectrum assignment.
- Bandwidth - expensive.
- FCC reports - inefficiency of spectrum usage.

- Dynamic Spectrum Access



COGNITIVE RADIO

- **Cognitive radio is an intelligent, aware, adaptive radio / system that senses its operational environment and dynamically adjust its radio operating parameters to modify system such as maximizing throughput, mitigating interference and facilitating interoperability.**

CHARACTERISTICS OF COGNITIVE RADIO

Cognitive capability

- Ability of CR to sense the radio environment by capturing variations and avoid the interferences.

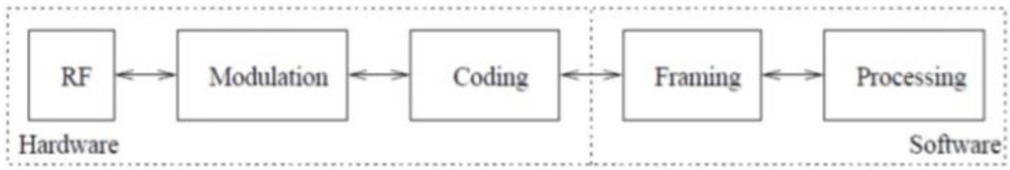
Reconfigurability

- Enables the CR to dynamically change the operating parameters according to the radio environment.

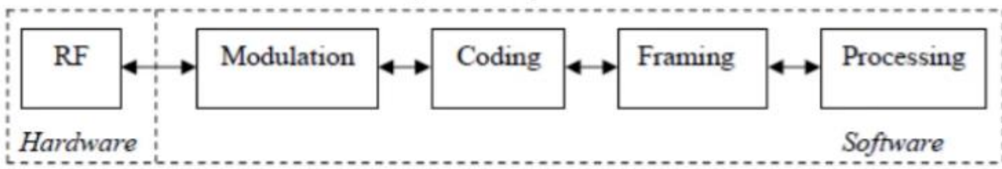
Achieving these characteristics in CR is well explained under the topic **"Cognition Cycle".**

COMPARISON OF RADIOS

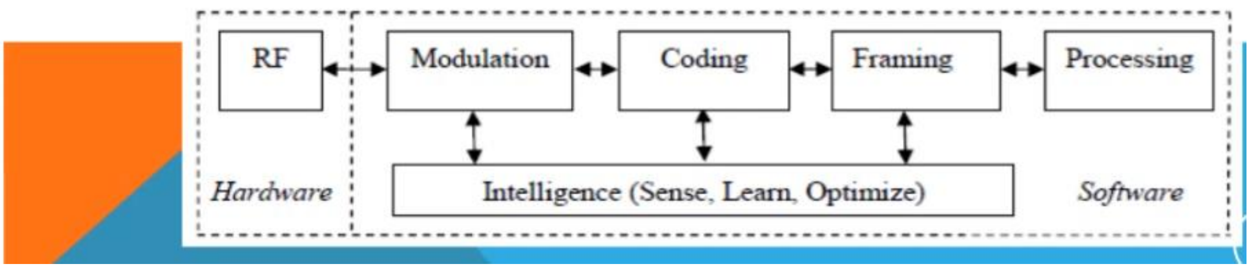
Traditional Radio:



Software Defined Radio(SDR):



Cognitive Radio(CR):



COGNITIVE RADIO - SMART & ALERT

- It knows where it is.
- It knows what services are available, for ex, it can identify and then use empty spectrum to communicate more efficiently.
- It knows what services interest the user, and knows how to find them.
- It knows the current degree of needs and future likelihood of needs of its user.
- Learns and recognizes usage patterns from the user.
- Applies "Model Based Reasoning" about user needs, local content and environmental context.

COGNITIVE RADIO – VARIETY OF CONTEXTS

Spectrum aware:

- Can autonomously exploit locally unused spectrum to provide new paths to spectrum access.

Policy Aware:

- Can roam across borders and adjust themselves to stay in compliance with local regulations.

Network Aware:

- Can negotiate with several service providers (networks) to connect a user at the lowest cost (or optimal performance).

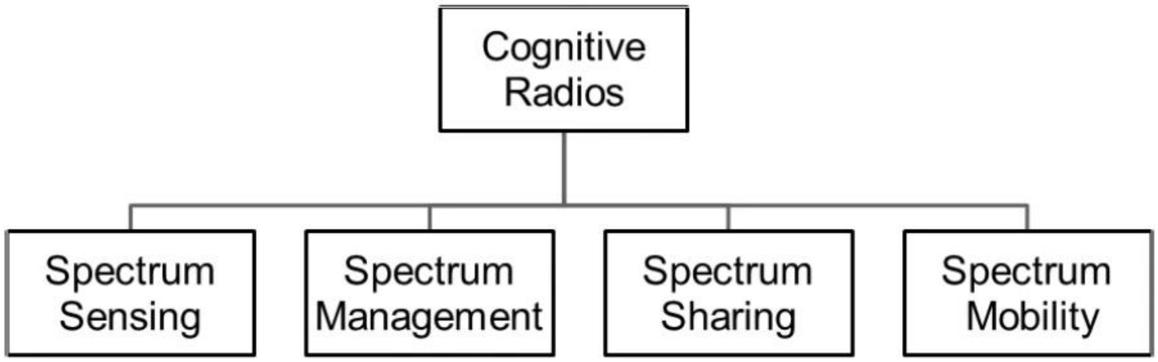
Channel Aware:

Can adapt themselves and their emissions without user intervention.

User Aware:

Can understand and follow actions and choices taken by their users and over time learn to become more responsive and to anticipate their needs.

Main functions of Cognitive Radio



Spectrum Sensing: Detect unused portion of spectrum or spectrum hole. (i.e) detect the presence / absence of the Primary user.

Spectrum Management: Capture the best spectrum to meet user communication requirements.

Spectrum Sharing: Provide fair spectrum scheduling method among CR users - distribute spectrum among the scheduled users.

Spectrum Mobility: Perform transition to a better spectrum, (i.e) vacate the channel, when licensed user is detected.